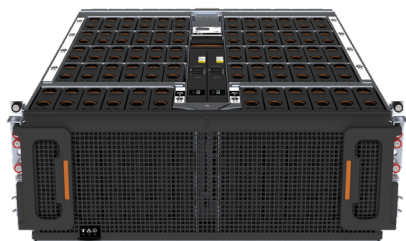


Ultrastar® Data60 3000 with Ceph Object Storage: The Scalable High Density Object Storage Platform



Challenges

Organizations dealing with AI, analytics, backup, and cloud-native workloads face multiple challenges:

- Explosive data growth (PB → EB scale) requires horizontal scalability.
- Traditional storage lacks elasticity and can become cost-prohibitive.
- All-flash is expensive for bulk storage.
- Need for high-capacity, low-cost storage tiers.
- Dense HDD enclosures can suffer from vibration and cooling inefficiencies.
- Performance degradation in large-scale deployments.

Solution

This solution integrates Ceph Tentacle Object Storage with Ultrastar Data60 3000 to deliver a scalable, high throughput object storage platform.

- Ceph's distributed, software defined architecture.
- Ultrastar Data60 3000 is a high-density, hybrid HDD-based storage platform.
- Modern erasure coding and replication policies optimized for object workloads.
- This results in a flexible object storage platform suitable for analytics, AI/ML data lakes, backup, archive, and cloud native applications.

Highlights

- Unified Software-Defined Storage: Ceph provides object, block, and file storage in a single platform.
- Validated performance across replication and EC (4+1, 4+2) policies with Ultrastar Data60 3000.

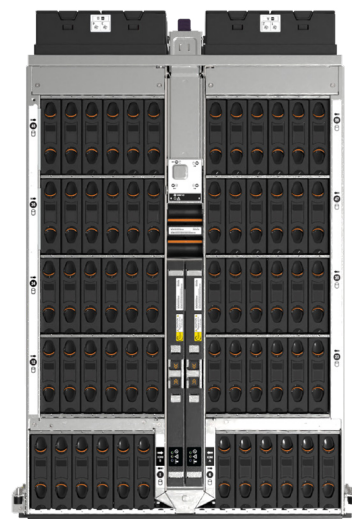
- Massive Scalability: Scale from few nodes to thousands, supporting petabytes to exabytes with Ultrastar Data60 3000.
- No Single Point of Failure: Distributed architecture with MON (cluster state), OSD (data storage), MGR (automation & monitoring), Intelligent Data Placement.
- CRUSH algorithm ensures: automatic data distribution, dynamic rebalancing without central bottlenecks
- Cloud-Native Object Storage: S3/Swift compatible via RADOS Gateway (RGW).
- Demonstrated balance between throughput, resilience, and storage efficiency.

Ultrastar Data60 3000 Storage Platforms

The Data60 3000 Hybrid Storage Platform is a key element of next-generation storage and software-defined storage (SDS) systems, delivering high density and the flexibility to balance performance with cost. The Ultrastar Data60 3000 Hybrid Storage Platform provides up to 1.92PB of raw storage in a compact and efficient form factor. Western Digital HelioSeal® drives ensure cool running, quiet operation, and high reliability. This platform addresses the demanding storage needs of large enterprise customers, storage OEMs, cloud service providers, and resellers/integrators that require dense, shared HDD storage. The Ultrastar Data60 3000 Hybrid Storage Platform provides the flexibility to specify the HDD combinations to balance capacity, performance, and cost. Developing storage devices and platforms side-by-side, we address these challenges through Silicon to Systems design, a set of technologies developed based on a holistic view of devices, platform, and their interactions. The first two of these innovative technologies are IsoVibe™ and ArcticFlow™. IsoVibe reduces vibration-induced performance degradation, while ArcticFlow overcomes the cooling issues by introducing cool air into the middle of the platform. Both these technologies contribute to long-term reliability, enabling our five-year limited warranty on the entire platform.

Features

- Up to 60 Ultrastar HDDs (SAS or SATA)
- Up to 1.92PB¹ of raw storage in 4U
- Up to four units may be daisy-chained for a total raw capacity of 7.68PB
- Patented IsoVibe technology helps ensure maximum performance even in heavy workloads
- Improved cooling from innovative ArcticFlow technology
- Choose dual-port SAS for high availability or single-port SATA for low cost
- Up to 12 x 24Gb/s SAS-4 host connections
- Enterprise-grade redundant and hot swappable PSUs, IO Modules, and fans
- Rack-mounted top cover for quick and easy service.



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Ceph Tentacle Object Storage

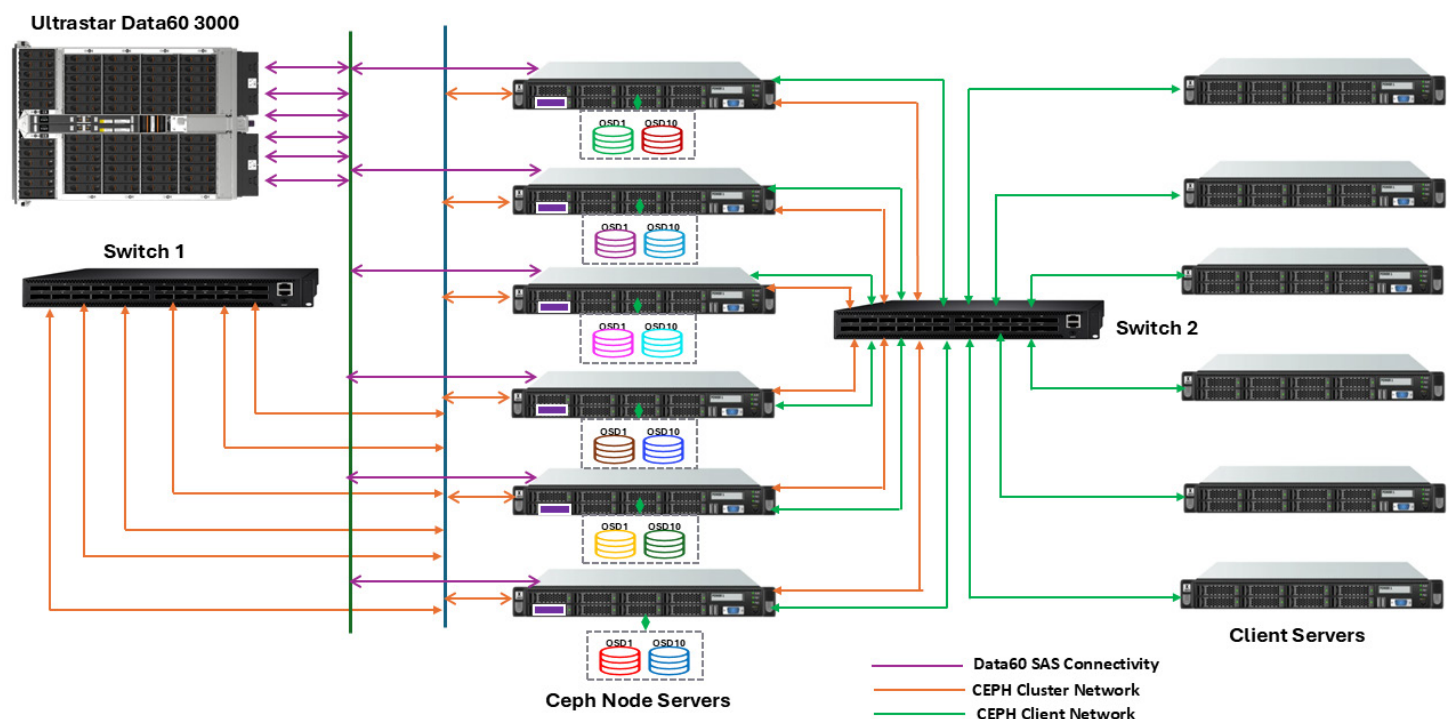
Ceph offers object storage tiering capabilities to optimize cost and performance by seamlessly moving data between storage classes. These tiers can be configured locally within an on-premises infrastructure or extended to include cloud-based storage classes, providing a flexible and scalable solution for diverse workloads. With policy-based automation, administrators can define lifecycle policies to migrate data between high-performance storage and cost-effective archival tiers, ensuring the right balance of speed, durability, and cost-efficiency. Ceph Tentacle introduces performance and efficiency improvements, particularly for erasure coded workloads.

- **CRUSH-based data placement** – Eliminates centralized metadata and enables deterministic, scalable, and highly resilient data distribution.
- **Massive horizontal scalability** – Scales from a few nodes to exabytes of object data without architectural changes.
- **Self-healing and self-managing** – Automatically detect failures and rebalances data with minimal administrative effort.
- **Native S3 and Swift APIs** – Provides industry standard object interfaces for seamless application integration.
- **Strong consistency model** – Ensures immediate data consistency, unlike many eventually consistent object stores.
- **Hardware and vendor agnostic** – Runs on commodity hardware and supports diverse storage media (HDD, SSD, NVMe™).
- **Integrated erasure coding and replication** – Offers flexible data protection schemes to balance durability, performance, and cost.
- **Multi-tenancy with quotas and isolation** – Enables secure sharing of object storage across teams or workloads.
- **Unified storage platform** – Object storage coexists with block and file storage under the same Ceph cluster.
- **Open-source and cloud-native** – Backed by a strong community, avoiding vendor lock in and supporting hybrid or private clouds. Ceph object storage provides S3 compatible access with strong consistency, self healing, and horizontal scalability.
- **Tentacle Innovations** – Offers Faster BlueStore WAL and improved compression, Optimized RGW bucket listing via faster OMAP iteration, Enhanced dashboard and multi-cluster management, Performance improvements for EC (FastEC).

Reference Architecture

The solution architecture combines Ceph Tentacle object storage deployed on industry standard servers tightly integrated with Ultrastar Data60 3000. Ultrastar Data60 3000 is a high density storage enclosure to deliver a scalable, resilient, and cost optimized storage. At the core, Ceph operates as a distributed, software defined storage layer built on RADOS, where OSDs (Object Storage Daemons) reside on HDDs within the Ultrastar Data60 3000 platform, while MON and MGR services run across a small set of control nodes to maintain cluster state and orchestration. Client access is provided through RADOS Gateway (RGW), exposing S3/Swift-compatible APIs for cloud-native applications. The architecture separates the public (client) network and cluster (replication) network to ensure optimal performance and fault isolation, while data is distributed intelligently using the CRUSH algorithm, eliminating centralized bottlenecks, and ensuring uniform utilization across the cluster. Each Ultrastar Data60 3000 enclosure contributes dense, high-capacity HDD-based OSD nodes, optionally augmented with SSDs for BlueStore WAL/DB acceleration, enabling a hybrid tiering approach where metadata and hot data benefit from flash performance while bulk object data resides on cost-efficient disks. This design allows seamless horizontal scaling by simply adding more UD60-backed nodes, delivering petabyte-to-exabyte growth with consistent performance, high availability, and self-healing capabilities, making it ideal for large-scale object storage, backup, and data lake workloads.

Ceph: Object Storage Deployment Diagram with Ultrastar Data60



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Performance Details

The measured performance numbers are specific for the current configuration of Ceph Tentacle with Ultrastar Data60 3000 topology in conjunction with the uses of 60 SATA HDDs/ osds with single SATA path connection via six storage nodes with 10*6 zoning configuration. One NVMe SSD is used in each storage server for the BlueStore WAL/DB acceleration. Warp tool is used to run the performance benchmarking on the system. The performance test is run using Warp, simultaneously running tests from all the 6 Ubuntu clients.

Ceph Object Storage Replication Performance with Ultrastar Data60 3000			Ceph Object Storage Erasure Coding Performance with Ultrastar Data60 3000		
Policy Details	Profile	Throughput	Policy Details	Profile	Throughput
Replication	GET	12.83 GBps	Erasure Coding	GET	6.30 GBps
(3-Copies)	PUT	2.03 GBps	(4+2)	PUT	2.66 GBps

Note: The performance results are captured after running multiple iterations with different data size (object size, objects, concurrency, policies).
Disclaimer: All the performance data are subject to change depending on the Drive model, capacity, workloads, servers, CPU, tunings and HA topology used.

Configurations Details

Metrics	Details
Storage Product	Ultrastar Data60 3000
Ceph Version	Ceph Tentacle version 20.2.1
Storage Interface	SATA
HBA Details	eHBA 9600-16e Tri-Mode Storage Adapter, FW: 8.17.1.0-00000-00001, Driver: 8.17.1.0.0
Server Details	PowerEdge® R550
Host OS	Ubuntu 22.04.05
Host NIC	CX6
Firmware Version	24.10-4.1.4 (22.32.2004 (MT_0000000437))
CPU	Intel® Xeon® Gold 5318Y CPU @ 2.10GHz or Higher CPU Configuration
CPU Core Details	Dual socket server with 24 core CPU each. 96 logical cores in total with HT enabled
Memory	256 GiB
Switch Details	NVIDIA® Onyx™ 3700
Drive Details	Ultrastar DC HC560 20TB CMR SATA Drives, WUH722020BLE604, PQGSWGC8 Sandisk® SN655 (7.68TB), WUS4BA176DSP3X4, R2210801 – one NVMe drive connected to each Storage server
Tool	Warp v1.4.0

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Advantages

High Density Meets Scale-Out Storage

- Ceph thrives on dense OSD layouts.
- Ultrastar Data60 3000 provides maximum capacity per rack unit.

Cost Efficiency (TCO Optimization)

HDD-based storage enables:

- Low cost per TB.
- Ideal for cold/warm object storage tiers.

Hybrid Flexibility

That Combines:

- HDD for capacity tiers.
- Local SSD for metadata and WAL/DB distribution for faster processing and performance.
- Aligns perfectly with Ceph BlueStore architecture.

Predictable Performance at Scale

- IsoVibe reduces vibration-induced latency and results in consistent OSD performance.
- Ultrastar Data60 3000 offers thermal stability with ArcticFlow.
- It ensures even cooling and reduces failure rates for better cluster stability

SDS Compatibility

Designed specifically for:

- Software-defined storage (SDS)
- Cloud, AI, and analytics workloads

Operational Simplicity with cephadm orchestration and automation

Supports:

- AI/ML pipelines
- Big data analytics
- Multi-site geo-replication

Conclusion

The combination of Ceph Tentacle Object Storage and Ultrastar Data60 3000 delivers a powerful, scalable, and cost-efficient storage solution for modern enterprises.

Ceph provides:

- Intelligent, self-healing distributed object storage.
- Cloud-native APIs and massive scalability.

Ultrastar Data60 3000 provides:

- High-density, reliable HDD infrastructure.
- Optimized performance through advanced engineering.

Together, they create a robust software-defined storage platform that:

- Maximizes storage density
- Minimizes cost
- Ensures enterprise-grade reliability

For organizations building petabyte-scale object storage, Ultrastar Data60 + Ceph Tentacle is an optimal architecture-balancing performance, capacity, cost, and operational simplicity.



¹One gigabyte (GB) is equal to one billion bytes and one petabyte (PB) is equal to one quadrillion bytes. Actual user capacity may be less due to operating environment.